The incandescent light bulb produces electric light through incandescence, which is the emission of light from an object that has been heated to a high temperature. In a typical incandescent bulb, a solid tungsten (W) filament, surrounded by a sealed glass bulb filled with an inert gas, is heated to a high temperature ($T_{filament} \approx 3300$ K) via an electric current. As the filament is heated, W(*s*) transitions to W(*g*). The vapor pressure of W(*g*), P_W, is related to the W(*s*) heating temperature, *T*, according to a modified version of the Clausius-Clapeyron equation (Equation 1).

$$P_W = 10^{7.78 - \frac{44680}{T}}$$

Equation 1

As W(*g*) is formed, it diffuses toward the cooler bulb wall ($T_{bulb\ wall} \approx 1000$ K). At this cooler temperature, W(*g*) transitions back to the solid state and accumulates on the bulb wall, which reduces the transmission of light and causes quick consumption of the filament.

To counteract this issue, a variation of the incandecent bulb, known as the tungsten-halogen bulb, was developed. In a tungsten-halogen bulb, a small amount of halogen gas $X_2(g)$ (X = Cl, Br, or I) is added to the inert gas. W(*g*) reacts with $X_2(g)$ only at the bulb wall to form transparent $WX_2(g)$ (Reaction 1).

$$W(g) + X_2(g) \rightleftarrows WX_2(g)$$

Reaction 1

$WX_2(g)$ then diffuses toward the hot filament, where it decomposes back into W(*g*) and $X_2(g)$. Some of the W(*g*) then reattaches to the filament, which allows the regenerative cycle to continue and prolongs the life of the bulb. A schematic of an operating tungsten-halogen bulb is shown in Figure 1.

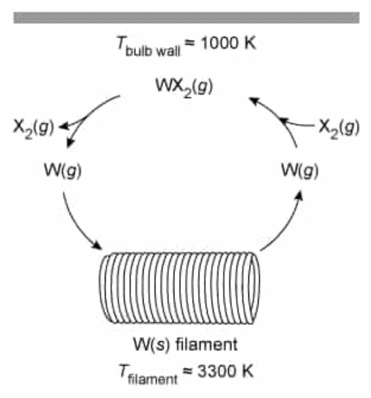


Figure 1 Schematic representation of an operating tungsten-halogen bulb (Note: X = Cl, Br, or I)

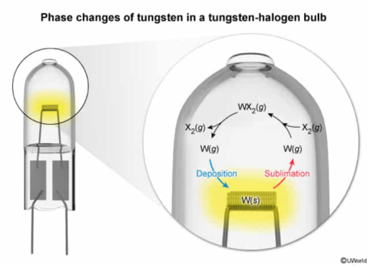
Tungsten-halogen bulbs emit ~10% of their energy as light and ~90% as heat, which causes a temperature increase in the surrounding environment.

Which of the following phase changes occur as the tungsten-halogen bulb operates?

- A. Vaporization and condensation [4%]
- B. Sublimation and freezing [2%]
- C. Vaporization and deposition [11%]
- ✔ D. Sublimation and deposition [82%]

Correct
82% answered correctly

Explanation:



A substance (ie, matter) can exist in one of three common phases (ie, states): gas, liquid, or solid. When enough energy (transferred as heat) is either absorbed or released by a substance, a physical process can occur in which the substance transitions from one phase to another (ie, a phase change). There are six types of phase changes that can occur between the solid, liquid, and gas states:

- Matter can transition from a gas to a liquid (**condensation**) or from a liquid to a gas (**vaporization**).
- Matter can transition from a liquid to a solid (**freezing**) or from a solid to a liquid (**melting**).
- Matter can transition from a solid to a gas (**sublimation**) or from a gas to a solid (**deposition**).

A tungsten-halogen bulb contains a W(*s*) filament, and when the filament is heated to a high temperature, W(*s*) is directly converted to W(*g*). After W(*g*) reacts with $X_2(g)$ to form $WX_2(g)$, $WX_2(g)$ decomposes near the hot filament to form W(*g*), which then reattaches to the W(*s*) filament. Therefore, based on the definitions above, both sublimation ($W(s) \rightarrow W(g)$) and deposition ($W(g) \rightarrow W(s)$) occur as the bulb operates.

(Choices A, B, and C) Solid tungsten atoms transition directly to the gas phase during heating, which is characterized as *sublimation*. Then, the gaseous tungsten atoms transition directly to the solid phase as they are *deposited* back onto the filament.

Educational objective:
Matter can exist in a solid, liquid, or gaseous state. Sublimation occurs when a substance in the solid state transitions directly to the gas state. Deposition occurs when a substance in the gas state transitions directly to the solid state.

Time spent:0 sec
Subject:
General Chemistry

QID:403817
System:
Thermodynamics, Kinetics & Gas Laws

The incandescent light bulb produces electric light through incandescence, which is the emission of light from an object that has been heated to a high temperature. In a typical incandescent bulb, a solid tungsten (W) filament, surrounded by a sealed glass bulb filled with an inert gas, is heated to a high temperature ($T_{filament}$ ≈ 3300 K) via an electric current. As the filament is heated, W(*s*) transitions to W(*g*). The vapor pressure of W(*g*), P_W, is related to the W(*s*) heating temperature, *T*, according to a modified version of the Clausius-Clapeyron equation (Equation 1).

$$P_{\mathrm{W}} = 10^{7.78 - \frac{44680}{T}}$$

Equation 1

As W(*g*) is formed, it diffuses toward the cooler bulb wall ($T_{bulb\ wall}$ ≈ 1000 K). At this cooler temperature, W(*g*) transitions back to the solid state and accumulates on the bulb wall, which reduces the transmission of light and causes quick consumption of the filament.

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$$\mathrm{W}(g) + \mathrm{X_2}(g) \rightleftarrows \mathrm{WX_2}(g)$$

Reaction 1

WX_2(*g*) then diffuses toward the hot filament, where it decomposes back into W(*g*) and X_2(*g*). Some of the W(*g*) then reattaches to the filament, which allows the regenerative cycle to continue and prolongs the life of the bulb. A schematic of an operating tungsten-halogen bulb is shown in Figure 1.

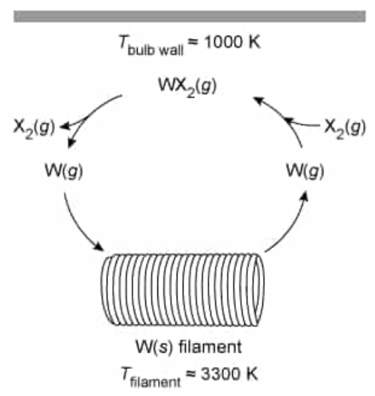
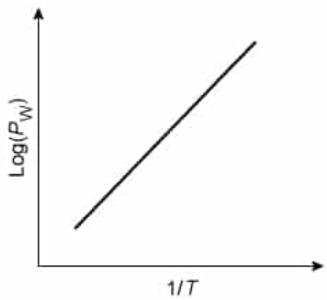


Figure 1 Schematic representation of an operating tungsten-halogen bulb (Note: X = Cl, Br, or I)

Tungsten-halogen bulbs emit ~10% of their energy as light and ~90% as heat, which causes a temperature increase in the surrounding environment.

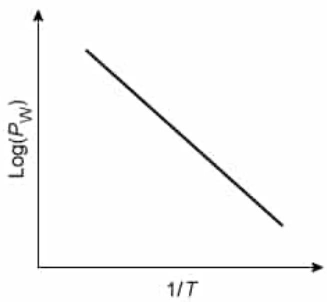
Which of the following graphs best illustrates the relationship between log(P_W) and 1/*T* in Equation 1?

○ A.



[25%]

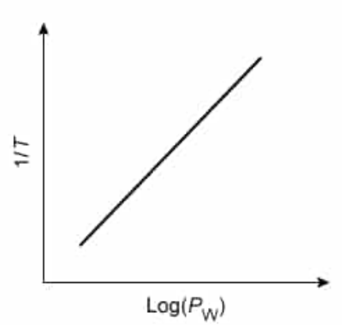
✔ ● B.



[59%]

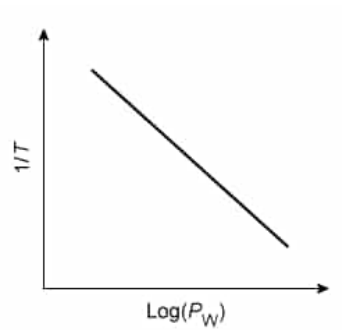
微信Mai_ange

○ C.



[6%]

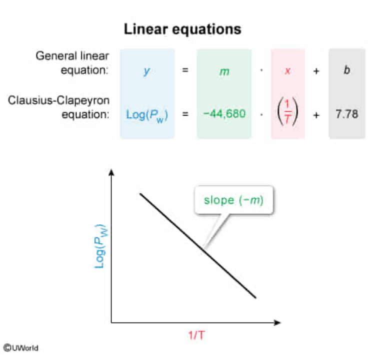
○ D.



[8%]

Correct
58% answered correctly

Explanation:



Graphs are used to show the relationship between a **dependent variable** (*y*-axis) and an **independent variable** (*x*-axis), both of which are measurable quantities. The value of the dependent variable depends on the independent variable; any change made to the independent variable will directly affect the dependent variable.

In this passage, the Clausius-Clapeyron equation (Equation 1) enables the evaluation of the partial pressure of tungsten P_W (dependent variable) at a temperature T (independent variable) and can be written as a linear function in the form of

$$y = \mathrm{mx} + b$$

where *x* and *y* are the *x*-axis and *y*-axis values, respectively; *m* is the slope of the line; and *b* is the *y*-intercept (ie, the point at which the line crosses the *y*-axis, when *x* = 0). In order to express Equation 1 as a linear function, the inverse logarithm (base 10) must first be removed by taking the logarithm of both sides of the equation:

$$\log(P_W) = \log\left(10^{7.78 - \frac{44680}{T}}\right)$$

$$\log(P_W) = 7.78 - \frac{44680}{T}$$

Then, rewriting the equation in the form of a linear function yields

$$\log(P_W) = -44680\left(\frac{1}{T}\right) + 7.78$$

where log of P_w corresponds to y, the reciprocal absolute temperature $1/T$ corresponds to x, and m corresponds to −44,680 (ie, a negative slope). Therefore, a graph of $\log(P_W)$ (y-axis) versus $1/T$ (x-axis) will show $\log(P_W)$ decreasing linearly with increasing $1/T$, as depicted in the graph shown in Choice B.

(Choice A) This graph shows $\log(P_W)$ increasing linearly with $1/T$ (*positive* slope) rather than decreasing with $1/T$ (*negative* slope).

(Choice C) This graph incorrectly shows $\log(P_W)$ increasing with increasing $1/T$ (positive slope). Additionally, because pressure is the *dependent* variable and temperature is the *independent* variable, pressure should be shown on the y-axis and temperature on the x-axis.

(Choice D) Although this graph correctly shows a line with a negative slope, pressure and temperature are plotted on the incorrect axes. Pressure is the *dependent* variable (y-axis), and temperature is the *independent* variable (x-axis).

Educational objective:
Equations can be expressed as linear functions in the form of $y = mx + b$, where y is the dependent variable, x is the independent variable, m is the slope of the line, and b is the y-intercept.

Time spent:0 sec
Subject:
General Chemistry

QID:403818
System:
Thermodynamics, Kinetics & Gas Laws

The incandescent light bulb produces electric light through incandescence, which is the emission of light from an object that has been heated to a high temperature. In a typical incandescent bulb, a solid tungsten (W) filament, surrounded by a sealed glass bulb filled with an inert gas, is heated to a high temperature ($T_{\text{filament}} \approx 3300$ K) via an electric current. As the filament is heated, W(*s*) transitions to W(*g*). The vapor pressure of W(*g*), P_W, is related to the W(*s*) heating temperature, *T*, according to a modified version of the Clausius-Clapeyron equation (Equation 1).

$$P_{\text{W}} = 10^{7.78 - \frac{44680}{T}}$$

Equation 1

As W(*g*) is formed, it diffuses toward the cooler bulb wall ($T_{\text{bulb wall}} \approx 1000$ K). At this cooler temperature, W(*g*) transitions back to the solid state and accumulates on the bulb wall, which reduces the transmission of light and causes quick consumption of the filament.

To counteract this issue, a variation of the incandescent bulb, known as the tungsten-halogen bulb, was developed. In a tungsten-halogen bulb, a small amount of halogen gas $X_2(g)$ (X = Cl, Br, or I) is added to the inert gas. W(*g*) reacts with $X_2(g)$ only at the bulb wall to form transparent $WX_2(g)$ (Reaction 1).

$$W(g) + X_2(g) \rightleftarrows WX_2(g)$$

Reaction 1

$WX_2(g)$ then diffuses toward the hot filament, where it decomposes back into W(*g*) and $X_2(g)$. Some of the W(*g*) then reattaches to the filament, which allows the regenerative cycle to continue and prolongs the life of the bulb. A schematic of an operating tungsten-halogen bulb is shown in Figure 1.

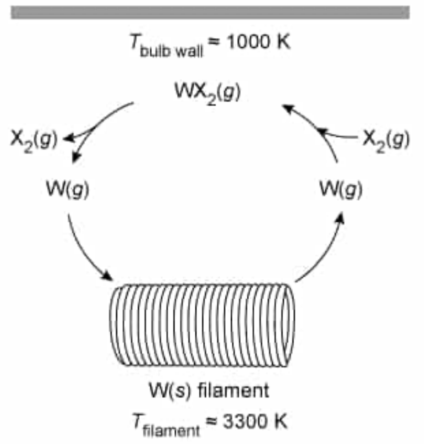


Figure 1 Schematic representation of an operating tungsten-halogen bulb (Note: X = Cl, Br, or I)

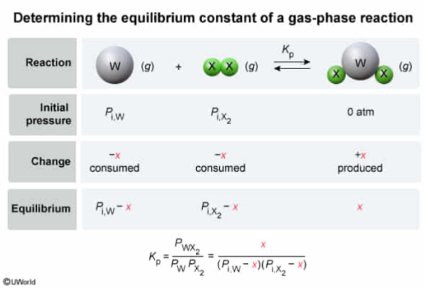
Tungsten-halogen bulbs emit ~10% of their energy as light and ~90% as heat, which causes a temperature increase in the surrounding environment.

Suppose W(*g*) has an initial partial pressure of $P_{\text{i, W}}$ and $X_2(g)$ has an initial partial pressure of $P_{\text{i, X}_2}$ near the walls of the bulb. Which of the following expressions is equivalent to K_p for Reaction 1 if the equilibrium partial pressure of $WX_2(g)$ is represented by *x*?

- A. $K_p = \dfrac{(P_{\text{i, W}} - x)(P_{\text{i, X}_2} - x)}{x}$ [11%]
- B. $K_p = \dfrac{-x}{(P_{\text{i, W}} + x)(P_{\text{i, X}_2} + x)}$ [4%]
- C. $K_p = \dfrac{(P_{\text{i, W}})(P_{\text{i, X}_2})}{x}$ [14%]
- ✔ D. $K_p = \dfrac{x}{(P_{\text{i, W}} - x)(P_{\text{i, X}_2} - x)}$ [70%]

Correct
70% answered correctly

Explanation:



For the generic gas-phase reaction between A(*g*) and B(*g*) to form products C(*g*), the equilibrium expression K_p can be written as a ratio of equilibrium partial pressures of products to reactants, each raised to the power of their balanced stoichiometric coefficients:

$$\mathrm{aA}(g) + \mathrm{bB}(g) \rightleftarrows \mathrm{cC}(g) \quad K_\mathrm{p} = \frac{(P_\mathrm{C})^\mathrm{c}}{(P_\mathrm{A})^\mathrm{a}(P_\mathrm{B})^\mathrm{b}}$$

In this passage, the gas-phase reaction occurring in the bulb has a corresponding K_p expression of

$$\mathrm{W}(g) + \mathrm{X_2}(g) \rightleftarrows \mathrm{WX_2}(g) \quad K_\mathrm{p} = \frac{P_\mathrm{WX_2}}{P_\mathrm{W}P_\mathrm{X_2}}$$

At the start of the reaction, the initial partial pressures of W(*g*) and X_2(*g*) near the bulb wall are defined as $P_\mathrm{i,W}$ and $P_\mathrm{i,X_2}$, respectively. As the reaction shifts toward equilibrium, W(*g*) and X_2(*g*) are depleted as WX_2(*g*) is formed. If the equilibrium partial pressure of WX_2(*g*) is defined as +*x*, then the equilibrium partial pressures of W(*g*) and X_2(*g*) must be equal to $P_\mathrm{i,W}$ − *x* and $P_\mathrm{i,X_2}$ − *x*, respectively, because they are both partially consumed. This can be demonstrated using an **ICE** (**I**nitial, **C**hange, **E**quilibrium) table:

	$\mathrm{W}(g)$	+	$\mathrm{X_2}(g)$	$\rightleftarrows$	$\mathrm{WX_2}(g)$
I	$P_\mathrm{i,W}$		$P_\mathrm{i,X_2}$		0 atm
C	$-x$		$-x$		$+x$
E	$P_\mathrm{i,W} - x$		$P_\mathrm{i,X_2} - x$		x

Therefore, substituting the equilibrium partial pressures into the K_p expression gives

$$K_p = \frac{x}{(P_\mathrm{i,W} - x)(P_\mathrm{i,X_2} - x)}$$

(Choice A) The equilibrium constant is defined as the ratio of products to reactants, *not* the ratio of reactants to products.

(Choice B) The reaction proceeds toward products, so W(*g*) and X_2(*g*) must decrease (−*x*) and WX_2(*g*) must increase (+*x*).

(Choice C) This expression incorrectly uses the *initial* partial pressures rather than the *equilibrium* partial pressures. Additionally, the ratio is inverted (ie, reactants to products instead of products to reactants).

Educational objective:
The gas-phase equilibrium constant can be expressed as

$$K_p = \frac{(P_\mathrm{products})^\mathrm{coefficients}}{(P_\mathrm{reactants})^\mathrm{coefficients}}$$

An ICE (Initial, Change, Equilibrium) table can be used to determine equilibrium partial pressures of reactants and products when initial conditions are given.

Time spent:0 sec
Subject:
General Chemistry

QID:403819
System:
Thermodynamics, Kinetics & Gas Laws

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$$P_W = 10^{7.78 - \frac{44680}{T}}$$

Equation 1

As W(*g*) is formed, it diffuses toward the cooler bulb wall ($T_{bulb\ wall}$ ≈ 1000 K). At this cooler temperature, W(*g*) transitions back to the solid state and accumulates on the bulb wall, which reduces the transmission of light and causes quick consumption of the filament.

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$$W(g) + X_2(g) \rightleftarrows WX_2(g)$$

Reaction 1

WX_2(*g*) then diffuses toward the hot filament, where it decomposes back into W(*g*) and X_2(*g*). Some of the W(*g*) then reattaches to the filament, which allows the regenerative cycle to continue and prolongs the life of the bulb. A schematic of an operating tungsten-halogen bulb is shown in Figure 1.

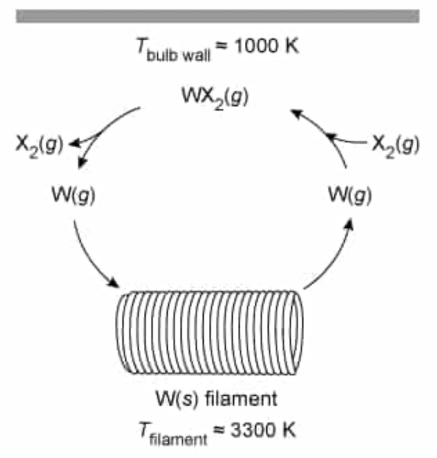


Figure 1 Schematic representation of an operating tungsten-halogen bulb (Note: X = Cl, Br, or I)

Tungsten-halogen bulbs emit ~10% of their energy as light and ~90% as heat, which causes a temperature increase in the surrounding environment.

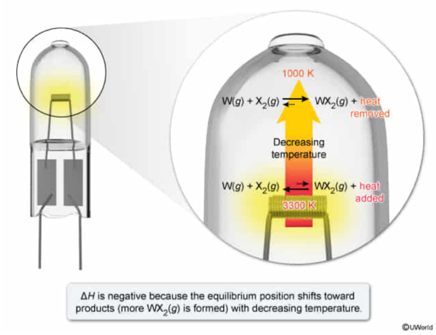
The enthalpy of reaction Δ*H* for Reaction 1 is best described by which of the following?

- A. Δ*H* is positive because the formation of WX_2(*g*) is favored with decreasing temperature. [12%]
- B. Δ*H* is positive because the formation of WX_2(*g*) is favored with increasing temperature. [25%]
- ✔ C. Δ*H* is negative because the formation of WX_2(*g*) is favored with decreasing temperature. [42%]
- D. Δ*H* is negative because the formation of WX_2(*g*) is favored with increasing temperature. [19%]

Correct
42% answered correctly

Explanation:

Enthalpy of reaction Δ*H* and Le Châtelier's principle



The amount of heat released or absorbed by a chemical reaction measured under constant pressure is equal to the change in enthalpy of the reaction (ΔH). Exothermic reactions release heat (ie, heat is a product), resulting in a **negative ΔH**, whereas endothermic reactions absorb heat (ie, heat is a reactant), resulting in a **positive ΔH**. For an exothermic reaction at equilibrium, a temperature decrease causes the reaction to shift toward products to compensate for the heat loss in accordance with Le Châtelier's principle. Conversely, for an endothermic reaction at equilibrium, a temperature decrease causes the reaction to shift toward reactants.

According to the passage, as the tungsten filament in the bulb is heated ($T_{filament} \approx 3300$ K), a temperature gradient forms between the hot filament and the cooler glass bulb wall ($T_{bulb\ wall} \approx 1000$ K). Near the bulb wall, W(*g*) reacts with X_2(*g*) to form WX_2(*g*) (Reaction 1). When WX_2(*g*) diffuses toward the hot filament, it decomposes back into W(*g*) and X_2(*g*). Therefore, the equilibrium favors the formation of WX_2(*g*) at lower temperatures and favors the formation of W(*g*) and X_2(*g*) at higher temperatures, indicating that the reaction is exothermic and **ΔH is negative**.

(Choices A, B, and D) Because the formation of WX_2(*g*) occurs at *lower* temperatures (bulb wall) and *not* at higher temperatures (filament), ΔH is *negative* for Reaction 1.

Educational objective:
For an exothermic reaction ($\Delta H < 0$), the equilibrium position shifts toward products with decreasing temperature. For an endothermic reaction ($\Delta H > 0$), the equilibrium position shifts toward reactants with decreasing temperature.

Time spent: 0 sec
Subject: General Chemistry

QID: 403820
System: Thermodynamics, Kinetics & Gas Laws

The incandescent light bulb produces electric light through incandescence, which is the emission of light from an object that has been heated to a high temperature. In a typical incandescent bulb, a solid tungsten (W) filament, surrounded by a sealed glass bulb filled with an inert gas, is heated to a high temperature ($T_{filament}$ ≈ 3300 K) via an electric current. As the filament is heated, W(*s*) transitions to W(*g*). The vapor pressure of W(*g*), P_W, is related to the W(*s*) heating temperature, *T*, according to a modified version of the Clausius-Clapeyron equation (Equation 1).

$$P_W = 10^{7.78 - \frac{44680}{T}}$$

Equation 1

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Reaction 1

$WX_2(g)$ then diffuses toward the hot filament, where it decomposes back into W(*g*) and $X_2(g)$. Some of the W(*g*) then reattaches to the filament, which allows the regenerative cycle to continue and prolongs the life of the bulb. A schematic of an operating tungsten-halogen bulb is shown in Figure 1.

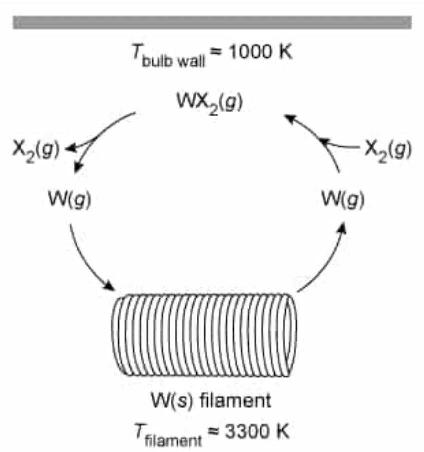


Figure 1 Schematic representation of an operating tungsten-halogen bulb (Note: X = Cl, Br, or I)

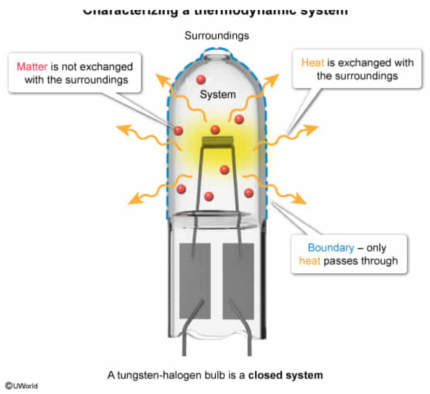
Tungsten-halogen bulbs emit ~10% of their energy as light and ~90% as heat, which causes a temperature increase in the surrounding environment.

An operating tungsten-halogen bulb is a(n):

- ✔ A. closed system, because only the heat radiated from the filament is exchanged with the surroundings. [59%]
- B. isolated system, because only the gas inside the bulb is exchanged with the surroundings. [3%]
- C. open system, because the heat radiated from the filament and gas inside the bulb are exchanged with the surroundings. [26%]
- D. isolated system, because the heat radiated from the filament and the gas inside the bulb are not exchanged with the surroundings. [10%]

Correct
60% answered correctly

Explanation:



A **thermodynamic system** can be defined as a space chosen for study that is physically separated by a boundary (eg, surface of a wall) from its **surroundings** (ie, the rest of the universe). The extent to which heat and matter can be transferred between a system and its surroundings depends on the type of separation barrier. As such, there are three common types of thermodynamic systems:

- **Open system**: Both heat and matter are freely exchanged with the surroundings.
- **Closed system**: Only heat is exchanged with the surroundings.
- **Isolated system**: Neither heat nor matter can be exchanged with the surroundings.

According to the passage, a tungsten-halogen bulb (ie, a thermodynamic system) emits light when the tungsten filament is heated (via an electric current) to very high temperatures. A large amount of heat is also radiated from the filament as the bulb operates. The bulb's glass wall (ie, the boundary) is a good conductor of heat and allows the heat to pass through freely into the surrounding air while the gas mixture (ie, matter) remains trapped in the bulb. As a result, the temperature of the surroundings increases. Based on the definitions given above, a tungsten-halogen bulb is best characterized as a **closed system** because **only heat** is exchanged with the surroundings.

(Choices B, C, and D) The heat radiated from the tungsten filament is freely exchanged with the surrounding air while the gas inside the bulb remains trapped. Therefore, a tungsten-halogen bulb is a *closed* system.

Educational objective:
A open system interacts with its surroundings through the transfer of both heat and matter, whereas a closed system interacts with its surroundings through the transfer of only heat. An isolated system does not interact with its surroundings.

Time spent:0 sec
Subject:
General Chemistry

QID:403821
System:
Thermodynamics, Kinetics & Gas Laws